THOMAS RICCI

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Education

Worcester Polytechnic Institute Expected Graduation: May 2027
BS in Computer Science
Relevant Coursework: Accelerated Introduction To Program Design, Accelerated Object-Oriented Design Concepts, Systems Programming Concepts, Algorithms, Databases, Computational Engineering, Computer Networks, Software Engineering, Operating Systems, Foundations of Computer Science, Theory of Computation, Distributed Computing Systems, Mobile & Ubiquitous Computing

Skills / Proficiencies

Proficient Languages: TypeScript & JavaScript, TSX & JSX, HTML, CSS, Nix, Java, Bash, Typst
Familiar Languages: Go, Rust, C, C++, Kotlin, Python, C#, SQL, R, Racket, MATLAB, Sass/SCSS, LaTeX
Libraries & Frameworks: React, Next.js, Node.js, Vite, TailwindCSS, ShadCN, Pixi.js, Prisma ORM, React Router, Tanstack, htmx, Bun, Express.js, OpenCV, NumPy, pandas, Cypress, Vitest, JUnit, Bootstrap
Tools & Platforms: Git, GitHub, GitHub Actions, Cloudflare (DNS, Proxy, Zero Trust, Workers, Pages, R2, Durable Objects, Tunnels & more), AWS, Gradle, Maven, Docker, Linux (NixOS, Ubuntu, Raspberry Pi OS & more), Figma, QEMU, Claude Code & other AI harnesses, VS Code, IntelliJ IDEA, Google Maps Platform, PostgreSQL, Oracle XE, nginx, Caddy, Jupyter Notebooks, ROS 2, Tailscale, Puppeteer, Playwright, ZFS

Work Experience

Invoice Cloud *June 2026 – Present*
Software Engineering Intern, Boston, MA
- Built an internal IAC platform that lets engineers create and manage repositories, Jira projects, and agent environments declaratively with code using TypeScript and Bun.
- Aided with development of an internal software factory for autonomous agentic development.
- Deploying these systems company-wide to over 3000 repos & 30 teams by EOY 2026.

The Intersection Company *Dec 2025 – Present*
Co-Founder and CTO, Remote
- Leading development of CiviCAD, a next-generation, cloud-native CAD software platform for roadway and corridor design that reduces time spent drawing plans by 62%.
- Building a highly-available, scalable, parametric CAD application with Cloudflare Workers, Durable Objects, R2, and React.
- Filed 2 provisional patents for a compliance-checking system that ensures roadways comply with the MUTCD, PDDG, STEP guidelines, Mass Amendments, and various other documents.
- Engaging in partnerships, research, and pilot programs with municipalities and universities.

Pyro, Inc. *Mar 2025 – Feb 2026*
Software Engineering Intern, Remote
- Designed and developed frontend experiences with Figma and Next.js.

Milton Academy *June 2025 – Sep 2025*
Web & Digital Support Intern, Remote
- Published web content, developed version control system for content managed by the Veracross CMS.

Xaverian Brothers High School IT Department *May 2022 – Aug 2023*
IT Technician, Westwood, MA

Projects

Humanoid Robot · anomalymotion.com *Aug 2026 – Present*

Go, Nix, Python, C++, JS, TS, React, Next.js, ShadCN, TailwindCSS, ROS 2, Cloudflare Zero Trust

- Building open-source humanoid robots in collaboration with other students at WPI.
- Maintaining a custom Linux distribution that consumes less than 512MB of memory and 1GB of storage for efficient operation on embedded devices.
- Leading development of a Next.js frontend platform allowing operators to manage robots.
- Maintaining self-hosted GitHub Actions runners and a Tailscale network for robot communication.

Hospital Management Platform *Mar 2025 – May 2025*

JS, TS, React, Vite, Node.js, Express.js, ShadCN, TailwindCSS, AWS, Docker, PostgreSQL, Prisma ORM

- Led team of 10 students developing hospital management software at WPI for the Mass General Brigham healthcare network using React and Express.js.
- Developed hospital navigation system with Google Maps Platform, a project management and ticketing system, employee directory, and employee message boards, storing data in a PostgreSQL database.
- Deployed this software with AWS EC2 and RDS and secured it behind the Cloudflare proxy.

FTC 19460 Captain & Director of Software · robotics.xbhs.com *Apr 2021 – Aug 2023*

Java, Kotlin, Gradle, Android, TS, JS, RoadRunner, OpenCV

- Founding member of FIRST® Tech Challenge competitive robotics team 19460 & led team to Massachusetts State Championship as Finalist Alliance in 2023.
- Developed a state machine to emulate a multi-threaded environment using one thread.
- Built intelligent autonomous routines using computer vision, SLAM algorithms, and odometry.
- Used reflection to design a system to manage execution of routines written in Java & Kotlin.

React Metaframework · octobox.thomasricci.dev *Nov 2021 – Mar 2023*

JS, TS, React, Vite, Node.js, ESLint, Stylelint, Puppeteer, TailwindCSS, SCSS

- Developed an SSG & CSR React metaframework similar to Next.js and Remix.
- Developed robust filesystem router on top of Tanstack Router (fka. React Location) with support for code splitting, lazy-loading, preloading and caching, parallel and waterfall loaders, route actions, and head management.
- Designed custom SSG pre-renderer using Puppeteer.
- Built a CLI wizard to bootstrap projects using this metaframework with support for ESLint, Stylelint, SCSS, Tailwind & a custom unit-testing framework.

Various Homelabbing Projects *June 2019 – Present*

Nix, Docker, Coolify, Pterodactyl, Pelican, Traefik, Caddy, nginx, JS, TS, Java, Python

- Deployed and monitored distributed workloads across a fleet of self-hosted machines using Nix.
- Deployed ZFS filesystems encrypted with AES-256-GCM to keep data secure.
- Managing internal networking, DNS, and a VPN for remote access with Cloudflare Zero Trust.
- Deploying and managing game servers, AI model inference, web services, tracking and analytics systems, and various other services.

Awards

Worcester Polytechnic Institute

- **Robert H. Grant Award**
  - Awarded for designing and prototyping a small, personal wind turbine for researchers in remote environments where weight and portability were key constraints that made both batteries and conventional electricity generation systems infeasible.